

Persistent Homological Decoding Optimization for High-Dimensional Quantum LDPC Codes

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Abstract

The current theoretical limitation in Quantum Error Correction (QEC) lies in the decoding efficiency of high-dimensional Quantum Low-Density Parity-Check (qLDPC) codes. While providing constant encoding rates and favorable distance scaling, existing decoding algorithms fail or become computationally intractable. We propose a framework introducing a localized filtration of the chain complex based on syndrome density to accelerate decoding, introducing a Persistent Syndrome Weight function to penalize error chains interacting with persistent topological features.

1 Introduction: The Global Information Gap

While surface codes (2D topological codes) provide a robust baseline for fault-tolerant quantum computation, their zero-rate scaling limits practical application. High-dimensional Quantum Low-Density Parity-Check (qLDPC) codes, specifically those derived from chain complexes of high-dimensional manifolds (e.g., 4D hyperbolic manifolds or lifted product codes), offer constant encoding rates and favorable distance scaling.

However, the critical gap lies in decoding efficiency. Existing decoding algorithms, such as Belief Propagation or Minimum Weight Perfect Matching (MWPM), fail or become computationally intractable ($O(n^3)$ or worse) when applied to the complex syndrome graphs of high-dimensional homological codes. This creates a bottleneck between theoretical code performance and practical fault tolerance.

2 Mathematical Formulation

Let a quantum CSS (Calderbank-Shor-Steane) code be defined by a chain complex of finite-dimensional vector spaces over $\mathbb{F}_2$:

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \quad (1)$$

where C_1 represents the qubit space, and the boundary operators ∂_1 and ∂_2^T define the X and Z stabilizer checks, satisfying $\partial_1 \circ \partial_2 = 0$.

The standard syndrome decoding problem seeks an error vector $e \in C_1$ given a syndrome $s \in C_0 \oplus C_2$, such that:

$$\partial_1 e_X = s_Z \quad \text{and} \quad \partial_2^T e_Z = s_X \quad (2)$$

minimizing the weight $wt(e)$.

The proposed framework introduces a localized filtration of the chain complex based on syndrome density to accelerate decoding. Define a filtered simplicial complex $\mathcal{K}_0 \subseteq \mathcal{K}_1 \subseteq \dots \subseteq \mathcal{K}_m = \mathcal{K}$ around the syndrome defects.

We introduce a ‘‘Persistent Syndrome Weight’’ function, $\mathcal{W}(e, \beta)$, which penalizes error chains not just by their Hamming weight, but by their interaction with the persistent topological features of the error graph. Let $\beta_k^{t,t+p}$ be the k -th persistent Betti number of the filtration.

The modified decoding objective becomes:

$$\arg \min_{e \in C_1} \left(\alpha \cdot wt(e) + \gamma \sum_{p=1}^m \Psi(\partial e, \beta_1^{t,t+p}) \right) \quad (3)$$

where Ψ is a penalty function increasing when the proposed error chain e forms homologically non-trivial cycles (logical errors) relative to the filtration steps, and α, γ are optimization parameters.

3 Theoretical Verification and Logic

The premise relies on mapping the decoding problem to a localized optimal bounding chain problem.

1. Standard MWPM: Solves the bounding chain problem in $O(|V|^3)$ for 2D graphs.

2. Persistent Filtration: By constructing a filtration based on syndrome proximity, we reduce the active graph size for matching.

3. Complexity Bound Formulation: If the filtration successfully bounds the search space such that the maximum cluster size processed is bounded by $c = \mathcal{O}(\log n)$, the localized matching complexity reduces to $\mathcal{O}(n \log^3 n)$.

4. Logical Consistency: The constraint $\partial_1 \circ \partial_2 = 0$ ensures that valid stabilizer generators do not trigger the penalty function Ψ , preserving the code’s structural integrity while penalizing logical errors during the localized search.

4 Conclusion

This formulation represents an exploratory synthesis combining Topological Data Analysis (persistent homology) with Quantum Information Theory (syndrome decoding). While the application of clustering algorithms (like Union-Find) to QEC is well-studied, dynamically integrating persistent Betti numbers as a penalty weight for high-dimensional qLDPC decoding opens a new pathway for the-

oretically efficient decoding in fault-tolerant quantum computation.